

Grade 11 Multi-Subject Practice Examination

Mathematics - Chemistry - Physics - Biology - English

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This booklet contains the answers, model responses, and brief explanations for the Grade 11 Multi-Subject Practice Examination.

Contents

Mathematics

Chemistry

Physics

Biology

English

Mathematics

Answer all questions in the spaces provided.

Choose the letter of the correct answer for each question.

1. $\log_2(8)$ is equal to:

- (A) 2
- (B) 3
- (C) 4
- (D) 8

Correct answer: (B) 3

Explanation: $8 = 2^3$, so $\log_2(8) = 3$.

2. Simplify: $3^4 \times 3^2$

- (A) 3^6
- (B) 3^8
- (C) 9^6
- (D) 3^2

Correct answer: (A) 3^6

Explanation: When multiplying powers with the same base, add the exponents: $4 + 2 = 6$, so the answer is 3^6 .

3. If $\sin(\theta) = 3/5$ and θ is an acute angle, what is $\cos(\theta)$?

- (A) $4/5$
- (B) $3/4$
- (C) $5/3$
- (D) $5/4$

Correct answer: (A) $4/5$

Explanation: Using a 3-4-5 right triangle, the adjacent side is 4, so $\cos(\theta) = 4/5$.

4. What is the next term in the arithmetic sequence 2, 5, 8, 11, ...?

- (A) 13
- (B) 14
- (C) 15
- (D) 16

Correct answer: (B) 14

Explanation: The common difference is 3, so the next term is $11 + 3 = 14$.

5. Find the sum of the first 5 terms of the geometric sequence 2, 6, 18, 54, ...

- (A) 121
- (B) 242
- (C) 243
- (D) 244

Correct answer: (B) 242

Explanation: $S_5 = 2(3^5 - 1)/(3 - 1) = 2(242)/2 = 242$.

6. What is the mean of the data set {4, 8, 6, 10, 12}?

- (A) 6
- (B) 7
- (C) 8
- (D) 9

Correct answer: (C) 8

Explanation: Sum = 40, and 40 divided by 5 values = 8.

7. A fair coin is tossed twice. What is the probability of getting exactly one head?

- (A) $\frac{1}{4}$
- (B) $\frac{1}{2}$
- (C) $\frac{3}{4}$
- (D) 1

Correct answer: (B) $\frac{1}{2}$

Explanation: Outcomes are HH, HT, TH, TT. Exactly one head occurs in 2 of the 4 outcomes, so $P = \frac{2}{4} = \frac{1}{2}$.

8. $\log_{10}(100)$ is equal to:

- (A) 1
- (B) 2
- (C) 10
- (D) 100

Correct answer: (B) 2

Explanation: $100 = 10^2$, so $\log_{10}(100) = 2$.

9. Solve for x: $\log_3(x) = 4$.

- (A) 12
- (B) 64
- (C) 81
- (D) 3

Correct answer: (C) 81

Explanation: $x = 3^4 = 81$.

10. The value of $\ln(e)$ (the natural logarithm of e) is:

- (A) 0
- (B) 1
- (C) e
- (D) 10

Correct answer: (B) 1

Explanation: By definition, $\ln(e) = \log_e(e) = 1$.

11. Which of the following is a term of the geometric sequence with $a_1 = 3$ and $r = 2$?

- (A) 8
- (B) 12
- (C) 14
- (D) 20

Correct answer: (B) 12

Explanation: The sequence is 3, 6, 12, 24, 48, ... so 12 is a term.

12. The variance of a data set is the square of its:

- (A) mean
- (B) median
- (C) mode
- (D) standard deviation

Correct answer: (D) standard deviation

Explanation: Variance = (standard deviation)².

13. How many different ways can 3 distinct books be arranged in a row on a shelf?

- (A) 3
- (B) 6
- (C) 9
- (D) 12

Correct answer: (B) 6

Explanation: There are $3! = 3 \times 2 \times 1 = 6$ possible arrangements.

14. If $P(A) = 0.3$ and $P(B) = 0.5$, and events A and B are mutually exclusive, what is $P(A \text{ or } B)$?

- (A) 0.15
- (B) 0.2
- (C) 0.8
- (D) 1.0

Correct answer: (C) 0.8

Explanation: For mutually exclusive events, $P(A \text{ or } B) = P(A) + P(B) = 0.3 + 0.5 = 0.8$.

15. Write 3^{-2} as a fraction.

- (A) -9
- (B) -1/9
- (C) 1/9
- (D) 9

Correct answer: (C) 1/9

Explanation: A negative exponent means reciprocal: $3^{-2} = 1/3^2 = 1/9$.

16. If $a_n = 2n + 1$, what is the 6th term of the sequence?

- (A) 11
- (B) 12
- (C) 13
- (D) 14

Correct answer: (C) 13

Explanation: $a_6 = 2(6) + 1 = 13$.

17. $2 \log(5)$ is equal to:

- (A) $\log(10)$
- (B) $\log(25)$
- (C) $\log(7)$
- (D) $\log(2.5)$

Correct answer: (B) $\log(25)$

Explanation: $2 \log(5) = \log(5^2) = \log(25)$.

18. The sum of the interior angles of a quadrilateral is:

- (A) 180
- (B) 270
- (C) 360
- (D) 450

Correct answer: (C) 360

Explanation: Any quadrilateral's interior angles sum to 360 degrees.

19. A fair six-sided die is rolled once. What is $P(\text{rolling an even number})$?

- (A) $1/6$
- (B) $1/3$
- (C) $1/2$
- (D) $2/3$

Correct answer: (C) $1/2$

Explanation: Even numbers are 2, 4, 6 - three of six outcomes, so $P = 3/6 = 1/2$.

20. Simplify $(2^3)^2$.

- (A) 2^5
- (B) 2^6
- (C) 2^8
- (D) 2^9

Correct answer: (B) 2^6

Explanation: When raising a power to a power, multiply exponents: $3 \times 2 = 6$.

21. An arithmetic sequence has first term 7 and common difference -2. What is the 4th term?

- (A) -1
- (B) 0
- (C) 1
- (D) 2

Correct answer: (C) 1

Explanation: $a_4 = 7 + 3(-2) = 7 - 6 = 1$.

22. In a class, test scores of 70 occur 8 times, 80 occurs 5 times, and 90 occurs 3 times. What is the mode?

- (A) 70
- (B) 80
- (C) 90
- (D) Cannot be determined

Correct answer: (A) 70

Explanation: The mode is the most frequently occurring value, and 70 occurs most often (8 times).

23. What is the square root of 100?

- (A) 10
- (B) 50
- (C) 100
- (D) 1000

Correct answer: (A) 10

Explanation: $10 \times 10 = 100$, so the square root of 100 is 10.

24. A standard deck of 52 playing cards has 26 red cards. What is the probability of drawing a red card?

- (A) $1/4$
- (B) $1/2$
- (C) $1/13$
- (D) $1/26$

Correct answer: (B) $1/2$

Explanation: $P(\text{red}) = 26/52 = 1/2$.

25. Which pair(s) of angles are complementary (sum to 90 degrees)?

- (A) 30 and 60 degrees
- (B) 45 and 45 degrees
- (C) Both of the above
- (D) 100 and 80 degrees

Correct answer: (C) Both of the above

Explanation: $30+60=90$ and $45+45=90$, so both pairs are complementary.

26. If the mode of a data set is 15, this means that:

- (A) 15 is the largest value
- (B) 15 occurs most frequently
- (C) 15 is the average of the data
- (D) 15 is the middle value

Correct answer: (B) 15 occurs most frequently

Explanation: The mode is defined as the value that occurs most often in a data set.

Circle T if the statement is true, or F if it is false.

27. T $\log(ab) = \log(a) + \log(b)$, for $a, b > 0$.

28. F $\sin(90 \text{ degrees}) = 0$.

29. F An arithmetic sequence has a constant common ratio between consecutive terms.

30. T Standard deviation is always a non-negative number.

31. T The probability of any event always lies between 0 and 1, inclusive.

32. T $\cos(0 \text{ degrees}) = 1$.

33. F In the geometric sequence 3, 6, 12, 24, the common ratio is 3.

34. F The median of a data set is always equal to its mean.

35. T The value of $\log(1)$ is 0 (for any valid base).

36. F Every arithmetic sequence is also a geometric sequence.

37. T Standard deviation measures how spread out data values are around the mean.

38. T The probability of two mutually exclusive events happening at the same time is always 0.

39. T $4!$ (4 factorial) is equal to 24.

40. T A sample space lists all the possible outcomes of a random experiment.

41. F The mode of a data set is always a single, unique value.

42. F $\log(0)$ is defined and equal to 0.

43. T The sum of the exterior angles of any convex polygon is always 360 degrees.

44. T A geometric sequence can have a negative common ratio.

45. T The range of a data set is the difference between its highest and lowest values.

46. F Combinations count the order of selection, while permutations do not.

47. T The probability of the complement of event A is $1 - P(A)$.

48. T $\log(a/b) = \log(a) - \log(b)$, for $a, b > 0$.

49. T The square root of 100 is 10.

50. **T** A standard deck of playing cards contains 52 cards, half of which are red.

51. **F** Two angles that sum to 90 degrees are called supplementary angles.

52. **T** The mode of a data set refers to the value that occurs most frequently.

Fill in each blank with the correct word, phrase, or value.

53. $\log_b(b) = \mathbf{1}$

54. The n th term of an arithmetic sequence is given by $a_n = a_1 + (n-1) \times \mathbf{d}$ (**the common difference**).

55. $\sin^2(\theta) + \cos^2(\theta) = \mathbf{1}$

56. The sum of the interior angles of a triangle is **180** degrees.

57. A sequence in which each term is found by multiplying the previous term by a fixed number is called a(n) **geometric** sequence.

58. The middle value of an ordered data set is called the **median**.

59. $5^0 = \mathbf{1}$

60. The probability of an impossible event is **0**.

61. Any nonzero number raised to the power 0 is equal to **1**.

62. $n!$ (n factorial) is the product of all positive integers from 1 up to **n** .

63. The average of the squared deviations of data values from the mean is called the **variance**.

64. $\log(1) = \mathbf{0}$ (for any valid base).

65. The sum of the first n terms of an arithmetic sequence is $S_n = (n/2)(2a_1 + (n-1)d)$.

66. The set of all possible outcomes of a random experiment is called the **sample space**.

67. Events that cannot both happen at the same time are called **mutually exclusive** events.

68. The square root of the variance is called the **standard deviation**.

69. The sum of the exterior angles of any convex polygon is always **360** degrees.

70. $\log(a/b) = \log(a) - \log(\mathbf{b})$.

71. An ordered arrangement of objects, where order matters, is called a(n) **permutation**.

72. A selection of objects where order does not matter is called a(n) **combination**.

73. The difference between the highest and lowest values in a data set is called the **range**.

74. $P(\text{not } A) = 1 - \mathbf{P(A)}$.

75. Two angles that sum to 90 degrees are called **complementary** angles.

76. Two angles that sum to 180 degrees are called **supplementary** angles.

77. The square root of a perfect square such as 100 is a whole, **rational** number.

78. A standard deck of playing cards contains **52** cards in total.

79. Table 1 - Standard trigonometric values (decimals rounded to 2 d.p. where needed)

Angle	sin	cos	tan
0 degrees	0.00	1.00	0.00

30 degrees	0.50	0.87	0.58
45 degrees	0.71	0.71	1.00
60 degrees	0.87	0.50	1.73
90 degrees	1.00	0.00	undefined

80. Table 2 - Frequency table of exam scores (25 students)

Score range	Frequency	Cumulative frequency
50-59	4	4
60-69	6	10
70-79	8	18
80-89	5	23
90-99	2	25

81. Table 3 - Converting between exponential and logarithmic form

Exponential form	Logarithmic form
$2^3 = 8$	$\log_2(8) = 3$
$10^2 = 100$	$\log_{10}(100) = 2$
$5^0 = 1$	$\log_5(1) = 0$

82. Classify each sequence below by putting a tick in the correct column.

Item	Arithmetic	Geometric	Neither
2, 4, 6, 8, 10, ...	X		
3, 6, 12, 24, ...		X	
1, 4, 9, 16, ...			X
100, 90, 80, 70, ...	X		
5, 15, 45, 135, ...		X	
2, 3, 5, 7, 11, ...			X
-8, -4, 0, 4, ...	X		
1, -2, 4, -8, ...		X	

Answer in one to three complete sentences.

83. Solve for x: $2^x = 32$. Show your working.

Model answer: $2^x = 2^5$, so $x = 5$.

84. Find the 10th term of the arithmetic sequence with $a_1 = 4$ and common difference $d = 3$.

Model answer: $a_{10} = 4 + 9(3) = 4 + 27 = 31$.

85. A die is rolled once. What is the probability of rolling a number greater than 4?

Model answer: Numbers greater than 4 are 5 and 6, so $P = 2/6 = 1/3$.

86. Convert the logarithmic equation $\log_5(25) = 2$ into exponential form.

Model answer: $5^2 = 25$.

87. Find the mean and the range of the data set: 12, 15, 9, 18, 21.

Model answer: Mean = $(12+15+9+18+21)/5 = 75/5 = 15$. Range = $21 - 9 = 12$.

88. Calculate $4!$ (4 factorial).

Model answer: $4! = 4 \times 3 \times 2 \times 1 = 24$.

89. A bag contains 3 red balls and 2 blue balls. One ball is drawn at random. What is the probability that it is red?

Model answer: $P(\text{red}) = 3/5$.

90. Find the value of x if $\log_2(x) = 5$.

Model answer: $x = 2^5 = 32$.

Write a well-organized, extended response of about 120-200 words.

91. A ball is dropped from a height of 5 metres. Each bounce reaches 60% of the height of the previous bounce. Explain how a geometric sequence can be used to model the height of each bounce, and calculate the height reached after the 4th bounce. Show all your working and reasoning.

Model answer / marking notes: The bounce heights form a geometric sequence with first term $a_1 = 5$ and common ratio $r = 0.6$, since each height is 0.6 times the previous one. Using $a_n = a_1 \times r^{(n-1)}$:
bounce 1 = $5 \times 0.6 = 3$ m, bounce 2 = 1.8 m, bounce 3 = 1.08 m, bounce 4 = 0.648 m (about 0.65 m).
A full-credit answer identifies the sequence type, states the formula, and shows each step of the calculation clearly.

92. Explain, using an example of each, the difference between an arithmetic sequence and a geometric sequence. Then describe one real-world situation that could be modelled by each type of sequence.

Model answer / marking notes: Open response. A strong answer defines an arithmetic sequence as one with a constant common difference between terms (e.g., 2, 5, 8, 11, ... with $d = 3$) and a geometric sequence as one with a constant common ratio between terms (e.g., 2, 6, 18, 54, ... with $r = 3$). Real-world examples might include a salary that increases by a fixed amount each year (arithmetic) versus a population or investment that grows by a fixed percentage each year (geometric). Marks are given for correct definitions, valid examples, and clear real-world applications.

93. Explain what standard deviation measures, and describe what it means for a data set of test scores to have a small standard deviation compared with a large standard deviation.

Model answer / marking notes: A strong answer explains that standard deviation measures how spread out (how much variation there is in) a set of data values around the mean. A small standard deviation means the test scores are clustered close to the mean (students performed consistently, with similar results), while a large standard deviation means the scores are spread widely above and below the mean (much more variability in performance between students).

94. Explain the difference between a permutation and a combination, and give one real-life example of when you would use each.

Model answer / marking notes: A strong answer explains that a permutation is an arrangement where order matters (e.g., ranking runners 1st, 2nd, and 3rd place in a race - swapping two runners' positions creates a different outcome), while a combination is a selection where order does not matter (e.g., choosing 3 students out of a class to form a group - the same three students form the same group regardless of the order they were picked).

Chemistry

Answer all questions in the spaces provided.

Choose the letter of the correct answer for each question.

1. Which of the following does NOT affect the rate of a chemical reaction?

- (A) Temperature
- (B) Concentration
- (C) A catalyst
- (D) The color of the container

Correct answer: (D) The color of the container

Explanation: Reaction rate depends on temperature, concentration, surface area, and catalysts - not the color of the container.

2. A catalyst speeds up a reaction by:

- (A) Increasing the temperature
- (B) Lowering the activation energy
- (C) Increasing the concentration
- (D) Increasing the pressure

Correct answer: (B) Lowering the activation energy

Explanation: A catalyst provides an alternative reaction pathway with a lower activation energy, without being consumed.

3. According to Le Chatelier's principle, increasing the pressure on a gaseous equilibrium system shifts the equilibrium toward the side with:

- (A) Fewer gas molecules
- (B) More gas molecules
- (C) No change occurs
- (D) Cannot be determined

Correct answer: (A) Fewer gas molecules

Explanation: The system shifts to counteract the change, favoring the side with fewer moles of gas to reduce the pressure increase.

4. A solution with pH = 3 is:

- (A) Strongly basic
- (B) Weakly basic
- (C) Neutral
- (D) Acidic

Correct answer: (D) Acidic

Explanation: Any pH below 7 (at 25 degrees C) indicates an acidic solution.

5. Which of the following is classified as a strong acid?

- (A) Acetic acid (CH_3COOH)
- (B) Hydrochloric acid (HCl)
- (C) Ammonia (NH_3)
- (D) Carbonic acid (H_2CO_3)

Correct answer: (B) Hydrochloric acid (HCl)

Explanation: HCl fully dissociates in water, making it a strong acid; the others are weak acids or a weak base.

6. In a redox reaction, oxidation is defined as:

- (A) Gain of electrons
- (B) Loss of electrons
- (C) Gain of protons
- (D) Loss of protons

Correct answer: (B) Loss of electrons

Explanation: Oxidation is the loss of electrons (remember: OIL RIG - Oxidation Is Loss, Reduction Is Gain).

7. How many moles are in 44 g of CO_2 ? (molar mass of $\text{CO}_2 = 44 \text{ g/mol}$)

- (A) 0.5
- (B) 1
- (C) 2
- (D) 4

Correct answer: (B) 1

Explanation: moles = mass / molar mass = $44 / 44 = 1$ mole.

8. At chemical equilibrium:

- (A) The reaction has completely stopped
- (B) The forward and reverse reaction rates are equal
- (C) Only the forward reaction occurs
- (D) Reactant and product concentrations are always equal

Correct answer: (B) The forward and reverse reaction rates are equal

Explanation: Equilibrium is dynamic: both reactions continue, but at equal rates, so concentrations remain constant (not necessarily equal to each other).

9. The molar mass of water (H_2O) is approximately:

- (A) 16 g/mol
- (B) 18 g/mol
- (C) 20 g/mol
- (D) 44 g/mol

Correct answer: (B) 18 g/mol

Explanation: $\text{H}_2\text{O} = 2(1) + 16 = 18 \text{ g/mol}$.

10. Which gas is typically produced when a reactive metal reacts with a dilute acid?

- (A) Oxygen
- (B) Hydrogen
- (C) Nitrogen
- (D) Carbon dioxide

Correct answer: (B) Hydrogen

Explanation: Metal + acid reactions typically produce a salt and hydrogen gas.

11. The pH scale commonly used in chemistry ranges from:

- (A) 0 to 7
- (B) 0 to 10
- (C) 0 to 14
- (D) 1 to 100

Correct answer: (C) 0 to 14

Explanation: The standard pH scale runs from 0 (most acidic) to 14 (most basic).

12. An exothermic reaction:

- (A) Absorbs heat from the surroundings
- (B) Releases heat to the surroundings
- (C) Has no energy change
- (D) Only occurs in solids

Correct answer: (B) Releases heat to the surroundings

Explanation: Exothermic reactions release energy (usually as heat) to the surroundings, often making them feel warm.

13. Which subatomic particle carries a negative electric charge?

- (A) Proton
- (B) Neutron
- (C) Electron
- (D) Nucleus

Correct answer: (C) Electron

Explanation: Electrons carry a negative charge and orbit the nucleus.

14. The process of a gas changing directly into a solid, without becoming a liquid, is called:

- (A) Evaporation
- (B) Sublimation
- (C) Deposition
- (D) Condensation

Correct answer: (C) Deposition

Explanation: Deposition is the direct gas-to-solid phase change (the reverse of sublimation).

15. Which statement best describes an isotope?

- (A) Atoms with a different number of protons
- (B) Atoms of the same element with different numbers of neutrons
- (C) Atoms with different numbers of electrons only
- (D) Molecules that share the same formula

Correct answer: (B) Atoms of the same element with different numbers of neutrons

Explanation: Isotopes share the same number of protons (same element) but differ in neutron number, giving different mass numbers.

16. Which type of bond involves atoms sharing pairs of electrons?

- (A) Ionic bond
- (B) Covalent bond
- (C) Metallic bond
- (D) Hydrogen bond

Correct answer: (B) Covalent bond

Explanation: A covalent bond forms when atoms share one or more pairs of electrons.

17. The chemical formula for sulfuric acid is:

- (A) H₂SO₃
- (B) H₂SO₄
- (C) HSO₄
- (D) SO₄

Correct answer: (B) H₂SO₄

Explanation: Sulfuric acid has the formula H₂SO₄.

18. Which of the following is an example of a physical change?

- (A) Burning paper
- (B) Melting wax
- (C) Rusting of a nail
- (D) Baking a cake

Correct answer: (B) Melting wax

Explanation: Melting is a change of state with no new substance formed, making it a physical change; the others involve chemical reactions.

19. How many valence electrons does a Group 1 (alkali metal) element typically have?

- (A) 1
- (B) 2
- (C) 7
- (D) 8

Correct answer: (A) 1

Explanation: Group 1 elements have a single electron in their outermost shell.

20. A pure substance made of only one type of atom is called a(n):

- (A) Compound
- (B) Mixture
- (C) Element
- (D) Solution

Correct answer: (C) Element

Explanation: An element consists of only one type of atom (one atomic number).

21. The technique used to separate a mixture based on differences in boiling point is called:

- (A) Filtration
- (B) Distillation
- (C) Chromatography
- (D) Decantation

Correct answer: (B) Distillation

Explanation: Distillation separates substances in a mixture by heating and condensing based on differing boiling points.

22. A solution that has dissolved as much solute as it possibly can at a given temperature is described as:

- (A) Dilute
- (B) Concentrated
- (C) Saturated
- (D) Supersaturated

Correct answer: (C) Saturated

Explanation: A saturated solution holds the maximum amount of dissolved solute possible at that temperature.

23. The chemical symbol for potassium is:

- (A) K
- (B) P
- (C) Po
- (D) Pt

Correct answer: (A) K

Explanation: Potassium's symbol, K, comes from its Latin name "kalium."

24. Which of these elements naturally exists as a two-atom (diatomic) molecule?

- (A) Helium
- (B) Neon
- (C) Oxygen
- (D) Argon

Correct answer: (C) Oxygen

Explanation: Oxygen naturally occurs as O₂, a diatomic molecule; the others are monatomic noble gases.

25. The process of a solid changing directly into a gas, without passing through the liquid state, is called:

- (A) Melting
- (B) Sublimation
- (C) Evaporation
- (D) Condensation

Correct answer: (B) Sublimation

Explanation: Sublimation is the direct solid-to-gas phase change.

26. Which subatomic particle carries no electric charge?

- (A) Proton
- (B) Electron
- (C) Neutron
- (D) Ion

Correct answer: (C) Neutron

Explanation: Neutrons are electrically neutral, unlike protons (positive) and electrons (negative).

Circle T if the statement is true, or F if it is false.

27. **T** Increasing the temperature of a reaction generally increases its rate.

28. **F** A catalyst is consumed during the reaction that it speeds up.

29. **T** A pH of 7 indicates a neutral solution at 25 degrees C.

30. **T** Adding more reactant to a system at equilibrium shifts the equilibrium toward the products.

31. **T** Bases turn red litmus paper blue.

32. **F** In electrolysis, reduction always occurs at the anode.

33. **F** Increasing the surface area of a solid reactant generally has no effect on reaction rate.

34. **T** A solution with pH 1 is more acidic than a solution with pH 5.

35. **T** Exothermic reactions release energy to the surroundings, usually as heat.

36. **F** All acids contain the element oxygen.

37. **T** Ionic bonds form through the transfer of electrons between atoms.

38. **F** Endothermic reactions release heat to the surroundings.

39. **T** Isotopes of an element have the same number of protons but different numbers of neutrons.

40. **F** A covalent bond involves the complete transfer of an electron from one atom to another.

41. **T** The pH of pure water at 25 degrees C is 7.

42. **T** Metal atoms typically lose electrons to form positively charged ions.

43. **T** A mixture can be separated by physical means, while a compound cannot be separated without a chemical reaction.

44. **F** Noble gases are generally very reactive elements.
45. **T** A concentrated solution contains a relatively large amount of dissolved solute.
46. **F** Filtration can be used to separate a dissolved (soluble) salt from water.
47. **T** The atomic number of an element equals the number of protons in its nucleus.
48. **T** All solutions are mixtures, but not all mixtures are solutions.
49. **T** The chemical symbol for potassium is K, from its Latin name "kalium."
50. **T** Oxygen naturally exists as a diatomic molecule, O₂.
51. **T** Sublimation is the change of state from solid directly to gas.
52. **F** Neutrons carry a positive electric charge.

Fill in each blank with the correct word, phrase, or value.

53. The branch of chemistry that studies the speed of reactions is called chemical **kinetics**.
54. A substance that increases the rate of a reaction without being permanently consumed is a(n) **catalyst**.
55. At equilibrium, the rate of the forward reaction equals the rate of the **reverse (backward)** reaction.
56. A solution with a pH less than 7 is described as **acidic**.
57. In the Bronsted-Lowry theory, a species that donates a proton (H⁺) is called a(n) **acid**.
58. In an electrochemical cell, oxidation occurs at the electrode called the **anode**.
59. One mole of any substance contains approximately **6.02 x 10²³ (Avogadro's number)** particles.
60. The reaction between an acid and a base that produces a salt and water is called a(n) **neutralization** reaction.
61. A reaction that absorbs heat energy from its surroundings is called **endothermic**.
62. The smallest particle of an element that retains its chemical properties is called a(n) **atom**.
63. Atoms that have gained or lost electrons, giving them an electric charge, are called **ions**.
64. The number of protons in the nucleus of an atom is called its **atomic** number.
65. A bond formed by the transfer of electrons from a metal to a non-metal is called a(n) **ionic** bond.
66. The substances present at the start of a chemical reaction are called the **reactants**.
67. The horizontal rows of the periodic table are called **periods**.
68. The vertical columns of the periodic table are called **groups** (or families).
69. A pure substance made of only one type of atom is called a(n) **element**.
70. A pure substance made of two or more elements chemically combined in fixed proportions is called a(n) **compound**.
71. The technique used to separate a mixture based on differences in boiling point is called **distillation**.
72. A solution containing the maximum amount of dissolved solute at a given temperature is described as **saturated**.
73. Elements in the same vertical column of the periodic table are called a **group**, and share similar chemical properties.

74. Elements in Group 18 of the periodic table are called the **noble** gases, and are generally unreactive.

75. The chemical symbol for potassium is **K**.

76. A change of state from solid directly to gas, without becoming a liquid, is called **sublimation**.

77. The subatomic particle with no electric charge is the **neutron**.

78. Oxygen naturally exists as a diatomic molecule with the formula **O₂**.

79. Table 1 - pH and the nature of a solution

pH value	Nature of solution	[H ⁺] compared with [OH ⁻]
2	acidic	[H ⁺] > [OH ⁻]
7	neutral	[H ⁺] = [OH ⁻]
11	basic	[H ⁺] < [OH ⁻]

80. Table 2 - Mole calculations (moles = mass / molar mass)

Substance	Molar mass (g/mol)	Mass given (g)	Moles
NaCl	58.5	117	2 mol
H ₂ O	18	36 g	2
CO ₂	44	22	0.5 mol

81. Table 3 - Concentration calculations (concentration = moles of solute / volume of solution in litres)

Moles of solute (mol)	Volume (L)	Concentration (mol/L)
0.5	1	0.5 mol/L
2 mol	0.5	4
1.5	3 L	0.5

82. For each change below, put a tick in the column that best describes it.

Item	Physical change	Chemical change
Melting ice	X	
Burning wood		X
Rusting of iron		X
Boiling water	X	
Digesting food		X
Dissolving sugar in water	X	
Photosynthesis in a leaf		X
Cutting a sheet of paper	X	

Answer in one to three complete sentences.

83. State two factors (other than a catalyst) that increase the rate of a chemical reaction.

Model answer: Any two of: increasing temperature, increasing concentration of reactants, increasing surface area of solid reactants, increasing pressure (for gases).

84. Calculate the number of moles in 9 g of water. (molar mass of H₂O = 18 g/mol)

Model answer: moles = 9 / 18 = 0.5 mol.

85. Write the word equation for the neutralization reaction between hydrochloric acid and sodium hydroxide.

Model answer: hydrochloric acid + sodium hydroxide → sodium chloride + water.

86. Explain why increasing pressure shifts a gaseous equilibrium toward the side with fewer moles of gas.

Model answer: By Le Chatelier's principle, the system responds to the added stress (higher pressure) by shifting in the direction that reduces the total number of gas molecules, which lowers the pressure back down.

87. A solution has a higher hydrogen ion concentration than a neutral solution. State whether it is acidic or basic, and give its approximate pH range.

Model answer: It is acidic; its pH is less than 7.

88. Write the balanced chemical equation for the reaction between magnesium metal and hydrochloric acid, producing magnesium chloride and hydrogen gas.

Model answer: $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$

89. Dissolving ammonium nitrate in water cools the resulting solution. State whether this process is exothermic or endothermic, and explain your reasoning.

Model answer: It is endothermic - the dissolving process absorbs heat energy from the surrounding water, which is why the temperature of the solution decreases.

90. An atom has 11 protons and 12 neutrons. State its atomic number and its mass number.

Model answer: Atomic number = 11 (the number of protons). Mass number = 11 + 12 = 23.

Write a well-organized, extended response of about 120-200 words.

91. Using collision theory, explain why increasing the temperature of a reaction increases its rate. Your answer should refer to particle motion, collision frequency, and the number of successful collisions.

Model answer / marking notes: A strong answer explains that raising the temperature increases the average kinetic energy of the particles, so they move faster. This increases both the frequency of collisions between reacting particles and, more importantly, the proportion of collisions that have enough energy to exceed the activation energy, resulting in more successful (effective) collisions per second and therefore a faster reaction rate.

92. Describe an experiment you could carry out to investigate how concentration of hydrochloric acid affects the rate of its reaction with magnesium ribbon. Include the variables you would control, the measurements you would take, and how you would present your results.

Model answer / marking notes: A good answer describes reacting fixed-length magnesium ribbon with HCl of varying concentrations (e.g., diluting a stock solution), keeping temperature, volume of acid, and ribbon length/mass constant (controlled variables), measuring the time taken for the ribbon to fully dissolve or the volume of hydrogen gas collected over time, repeating each concentration for reliability, and presenting the results in a table and a graph of rate (1/time) against concentration.

93. Explain the difference between ionic bonding and covalent bonding, giving one example compound formed by each type of bond.

Model answer / marking notes: A strong answer explains that ionic bonding involves the transfer of one or more electrons from a metal atom to a non-metal atom, producing oppositely charged ions that attract each other electrostatically (e.g., sodium chloride, NaCl), while covalent bonding involves atoms - usually non-metals - sharing one or more pairs of electrons to complete their outer electron shells (e.g., water, H₂O, or carbon dioxide, CO₂).

94. Explain the difference between a mixture and a compound, using an example of each, and describe one physical method that could separate the components of your example mixture.

Model answer / marking notes: A strong answer explains that a compound is formed when elements chemically bond together in a fixed ratio and can only be broken down by a chemical reaction (e.g., water, H₂O), while a mixture consists of two or more substances that are only physically combined, in variable proportions, and can be separated by physical methods (e.g., salt water can be separated by evaporation, or sand and iron filings can be separated using a magnet).

Physics

Answer all questions in the spaces provided.

Choose the letter of the correct answer for each question.

1. Ohm's Law states that:

- (A) $V = I/R$
- (B) $V = IR$
- (C) $V = R/I$
- (D) $I = VR$

Correct answer: (B) $V = IR$

Explanation: Ohm's Law: $V = IR$, where V is voltage, I is current, and R is resistance.

2. The SI unit of electric current is the:

- (A) Volt
- (B) Ohm
- (C) Ampere
- (D) Watt

Correct answer: (C) Ampere

Explanation: Electric current is measured in amperes (A).

3. Two resistors of 4 ohm and 6 ohm are connected in series. What is the total resistance?

- (A) 2 ohm
- (B) 10 ohm
- (C) 24 ohm
- (D) 1.5 ohm

Correct answer: (B) 10 ohm

Explanation: In series, resistances simply add: $4 + 6 = 10$ ohm.

4. In a parallel circuit, the voltage across each branch is:

- (A) Different for each branch
- (B) The same across each branch
- (C) Always zero
- (D) Equal to the total current

Correct answer: (B) The same across each branch

Explanation: Each branch of a parallel circuit is connected across the same two points, so they share the same voltage.

5. A converging (convex) lens forms a real, inverted image when the object is placed:

- (A) Inside the focal length
- (B) Exactly at the focal point
- (C) Beyond the focal length
- (D) Only at infinity

Correct answer: (C) Beyond the focal length

Explanation: When an object is beyond the focal length of a convex lens, the lens forms a real, inverted image.

6. Heat transfer through direct contact between vibrating particles is called:

- (A) Convection
- (B) Radiation
- (C) Conduction
- (D) Evaporation

Correct answer: (C) Conduction

Explanation: Conduction transfers thermal energy through particle-to-particle contact, without the particles themselves moving from place to place.

7. The SI unit of electrical power is the:

- (A) Ampere
- (B) Volt
- (C) Watt
- (D) Coulomb

Correct answer: (C) Watt

Explanation: Power is measured in watts (W), where 1 W = 1 joule per second.

8. Which of these materials is the best electrical conductor?

- (A) Rubber
- (B) Glass
- (C) Copper
- (D) Wood

Correct answer: (C) Copper

Explanation: Copper is a metal with free electrons and is an excellent electrical conductor; the others are insulators.

9. Specific heat capacity is defined as the amount of heat needed to raise the temperature of 1 kg of a substance by:

- (A) 1 degree Celsius
- (B) 10 degrees Celsius
- (C) 100 degrees Celsius
- (D) 0.5 degree Celsius

Correct answer: (A) 1 degree Celsius

Explanation: Specific heat capacity is defined per 1 degree Celsius (equivalently 1 Kelvin) rise in temperature, per kilogram.

10. A convex (diverging) mirror always produces a(n):

- (A) real, magnified image
- (B) virtual, diminished image
- (C) virtual, magnified image
- (D) real, diminished image

Correct answer: (B) virtual, diminished image

Explanation: A convex mirror always forms a virtual, upright, and diminished (smaller) image, regardless of object distance.

11. Two 10 ohm resistors are connected in parallel. What is their combined resistance?

- (A) 5 ohm
- (B) 10 ohm
- (C) 20 ohm
- (D) 50 ohm

Correct answer: (A) 5 ohm

Explanation: For two equal resistors in parallel: $R = (R_1 \times R_2)/(R_1 + R_2) = 100/20 = 5$ ohm.

12. Which of the following quantities is measured in joules?

- (A) Power
- (B) Current
- (C) Energy
- (D) Resistance

Correct answer: (C) Energy

Explanation: Energy (and work) is measured in joules (J); power is in watts, current in amperes, resistance in ohms.

13. The process of using a changing magnetic field to generate an electric current is called:

- (A) Magnetization
- (B) Electromagnetic induction
- (C) Conduction
- (D) Polarization

Correct answer: (B) Electromagnetic induction

Explanation: Electromagnetic induction is the process by which a changing magnetic field induces an electric current in a conductor.

14. A transformer that increases voltage (while decreasing current) is called a:

- (A) Step-down transformer
- (B) Step-up transformer
- (C) Isolation transformer
- (D) Rectifier

Correct answer: (B) Step-up transformer

Explanation: A step-up transformer increases the output voltage relative to the input voltage.

15. Which method of heat transfer can occur through the vacuum of space?

- (A) Conduction only
- (B) Convection only
- (C) Radiation only
- (D) All three equally

Correct answer: (C) Radiation only

Explanation: Radiation does not require a medium and can travel through a vacuum, unlike conduction and convection.

16. Which formula correctly gives the electrical energy transferred, E, in terms of power P and time t?

- (A) $E = P/t$
- (B) $E = P \times t$
- (C) $E = P + t$
- (D) $E = P - t$

Correct answer: (B) $E = P \times t$

Explanation: Energy transferred equals power multiplied by the time for which it is used: $E = Pt$.

17. The formula for kinetic energy is:

- (A) $KE = mgh$
- (B) $KE = (1/2)mv^2$
- (C) $KE = Fd$
- (D) $KE = P/t$

Correct answer: (B) $KE = (1/2)mv^2$

Explanation: Kinetic energy depends on mass and the square of velocity: $KE = (1/2)mv^2$.

18. The formula for gravitational potential energy is:

- (A) $PE = mgh$
- (B) $PE = (1/2)mv^2$
- (C) $PE = Fd$
- (D) $PE = IV$

Correct answer: (A) $PE = mgh$

Explanation: Gravitational potential energy: $PE = mgh$ (mass x gravitational field strength x height).

19. According to the law of conservation of energy, energy:

- (A) can be created but not destroyed
- (B) can be destroyed but not created
- (C) cannot be created or destroyed, only transformed
- (D) is always lost as heat

Correct answer: (C) cannot be created or destroyed, only transformed

Explanation: Energy is never created or destroyed - it only changes from one form to another.

20. For a wave travelling at constant speed, a higher frequency corresponds to a:

- (A) longer wavelength
- (B) shorter wavelength
- (C) identical wavelength
- (D) wave with no wavelength

Correct answer: (B) shorter wavelength

Explanation: Since $v = f \times \text{wavelength}$ is constant, wavelength and frequency are inversely related.

21. Which of the following is a vector quantity?

- (A) Speed
- (B) Distance
- (C) Mass
- (D) Velocity

Correct answer: (D) Velocity

Explanation: Velocity has both magnitude and direction, making it a vector quantity, unlike speed, distance, or mass.

22. The efficiency of a machine is calculated as:

- (A) $\text{useful output energy} / \text{total input energy} \times 100\%$
- (B) $\text{total input energy} / \text{useful output energy} \times 100\%$
- (C) output energy minus input energy
- (D) input energy plus output energy

Correct answer: (A) $\text{useful output energy} / \text{total input energy} \times 100\%$

Explanation: Efficiency compares how much of the input energy is converted into useful output energy.

23. The SI unit of frequency is the:

- (A) metre
- (B) second
- (C) hertz
- (D) newton

Correct answer: (C) hertz

Explanation: Frequency is measured in hertz (Hz), the number of cycles per second.

24. Which statement best describes an electrical insulator?

- (A) It allows electric current to flow through it easily
- (B) It does not allow electric current to flow through it easily
- (C) It is always magnetic
- (D) It is always transparent

Correct answer: (B) It does not allow electric current to flow through it easily

Explanation: Insulators strongly resist the flow of electric current.

25. Sound is best classified as a:

- (A) transverse wave
- (B) longitudinal wave
- (C) electromagnetic wave
- (D) standing wave only

Correct answer: (B) longitudinal wave

Explanation: Sound waves are longitudinal - particles vibrate parallel to the direction the wave travels.

26. Which formula correctly relates speed, distance, and time?

- (A) speed = distance x time
- (B) speed = distance / time
- (C) speed = time / distance
- (D) distance = speed + time

Correct answer: (B) speed = distance / time

Explanation: Speed is defined as distance travelled divided by the time taken.

Circle T if the statement is true, or F if it is false.

27. **T** Current flows more easily through a conductor with low resistance.

28. **T** In a series circuit, the current is the same at every point in the circuit.

29. **F** A concave mirror always produces a virtual image, no matter where the object is placed.

30. **T** Electrical power can be calculated using $P = VI$ (voltage multiplied by current).

31. **F** Radiation requires a medium (such as air) in order to transfer heat energy.

32. **T** Adding more resistors in parallel decreases the total resistance of a circuit.

33. **F** The north pole of a magnet repels the south pole of another magnet.

34. **T** Light travels more slowly in glass than it does in air.

35. **T** A step-up transformer increases voltage while decreasing current (assuming no energy losses).

36. **F** Specific heat capacity is the energy needed to change a substance's state without changing its temperature.

37. **T** Electromagnetic induction can generate an electric current by moving a magnet near a coil of wire.

38. **F** All metals have exactly the same specific heat capacity as each other.

39. **T** A convex mirror always forms a virtual, upright, diminished image, regardless of object distance.

40. **T** Electrical energy transferred can be calculated using $E = Pt$, where P is power and t is time.

41. **F** Sound waves are able to travel through a vacuum.

42. **T** Total internal reflection can occur when light travels from a denser medium into a less dense medium at a large enough angle.

43. **T** Kinetic energy depends on both the mass and the velocity of a moving object.

44. **T** Gravitational potential energy depends on an object's height above a reference point.
45. **T** At this level, energy is described as something that cannot be created or destroyed, only transformed.
46. **F** Velocity and speed are exactly the same physical quantity.
47. **T** No real machine can be 100% efficient, because some energy is always lost, often as heat.
48. **F** For a wave travelling at constant speed, wavelength and frequency are directly proportional to each other.
49. **T** The SI unit of frequency is the hertz (Hz).
50. **F** An electrical insulator allows current to flow through it easily.
51. **T** Sound waves are longitudinal waves.
52. **T** Speed is calculated by dividing distance by time.

Fill in each blank with the correct word, phrase, or value.

53. The SI unit of electrical resistance is the **ohm**.
54. A component that converts electrical energy into light energy is called a **lamp / light bulb**.
55. The transfer of heat through a fluid due to differences in density is called **convection**.
56. Electrical power can be written as $P = I^2 \times R$ (**resistance**).
57. A lens that is thinner in the middle than at the edges is called a **diverging (concave)** lens.
58. The magnetic field lines around a straight current-carrying wire form concentric **circles**.
59. The total resistance of resistors connected in series is found by **adding** the individual resistances.
60. The temperature at which a substance changes from a solid to a liquid is called its **melting** point.
61. The heat energy needed to raise the temperature of 1 kg of a substance by 1 degree C is called its **specific heat capacity**.
62. The energy needed to change a substance from solid to liquid without changing its temperature is called **latent** heat.
63. A device that uses electromagnetic induction to generate electricity is called a(n) **generator (dynamo)**.
64. A transformer that decreases voltage is called a(n) **step-down** transformer.
65. The bending of light as it passes from one transparent medium into another is called **refraction**.
66. The unit used on electricity bills to measure energy transferred is the kilowatt-**hour**.
67. The formula relating energy, power, and time is $E = P \times t$ (**time**).
68. A mirror that curves outward, like the back of a spoon, is called a **convex** mirror.
69. The formula for kinetic energy is $KE = (1/2) \times m \times v^2$ (**velocity squared**).
70. The formula for gravitational potential energy is $PE = m \times g \times h$ (**height**).
71. The law of conservation of energy states that energy cannot be created or **destroyed**, only transformed.
72. A quantity that has both magnitude and direction is called a **vector** quantity.

73. The efficiency of a machine is the ratio of useful output energy to **total** input energy, as a percentage.
74. For a wave travelling at constant speed, wavelength and frequency are **inversely** proportional to each other.
75. The SI unit of frequency is the **hertz**.
76. A material that does not allow electric current to flow through it easily is called a(n) **insulator**.
77. Sound waves are an example of a **longitudinal** wave.
78. The formula for speed is distance divided by **time**.

79. Table 1 - Ohm's Law calculations ($V = IR$)

Voltage (V)	Current (A)	Resistance (ohm)
12	2	6 ohm
10 V	5	2
9	3 A	3

80. Table 2 - Combining two resistors R1 and R2

R1 (ohm)	R2 (ohm)	Series total ($R1+R2$)	Parallel total ($R1R2/(R1+R2)$)
4	4	8 ohm	2 ohm
6	3	9	2 ohm
10	10 ohm	20	5

81. Table 3 - Kinetic energy calculations ($KE = 0.5 \times m \times v^2$)

Mass (kg)	Velocity (m/s)	Kinetic energy (J)
2	3	9 J
4 kg	5	50
1	4 m/s	8

82. For each material below, put a tick in the column that applies.

Item	Conductor	Insulator
Copper wire	X	
Rubber glove		X
Glass rod		X
Aluminium foil	X	
Plastic ruler		X
Iron nail	X	

Wooden block		X
Silver spoon	X	

Answer in one to three complete sentences.

83. Calculate the current through a 6 ohm resistor connected to a 12 V battery.

Model answer: $I = V/R = 12/6 = 2 \text{ A}$.

84. Two resistors of 5 ohm and 20 ohm are connected in parallel. Calculate the total resistance.

Model answer: $1/R = 1/5 + 1/20 = 4/20 + 1/20 = 5/20 = 1/4$, so $R = 4 \text{ ohm}$.

85. State one similarity and one difference between conduction and convection.

Model answer: Similarity: both transfer thermal energy from a hotter region to a cooler region.

Difference: conduction happens through direct particle-to-particle contact with no bulk movement of matter, while convection involves the bulk movement of a heated fluid (liquid or gas).

86. A converging lens has a focal length of 10 cm. An object is placed 25 cm from the lens (more than twice the focal length). State whether the resulting image is real or virtual, and whether it is magnified or diminished.

Model answer: The image is real and diminished (smaller than the object), because the object distance is greater than twice the focal length, placing the image between one and two focal lengths on the other side of the lens.

87. Explain why birds can safely perch on a high-voltage power line without being electrocuted.

Model answer: Both of the bird's feet touch the same wire, so there is almost no potential difference between them. Since current requires a difference in potential (a completed circuit between two different potentials, such as the wire and the ground), negligible current flows through the bird's body.

88. Calculate the energy transferred by a 2000 W kettle running for 3 minutes (180 seconds).

Model answer: $E = Pt = 2000 \times 180 = 360,000 \text{ J}$ (360 kJ).

89. State two differences between the images produced by a convex mirror and a concave mirror.

Model answer: A convex mirror always produces a virtual, upright, diminished image no matter where the object is placed, while a concave mirror can produce either a real, inverted image (object beyond the focal point) or a virtual, upright, magnified image (object within the focal point).

90. 500 J of heat energy raises the temperature of 2 kg of a substance by 5 degrees C. Calculate its specific heat capacity using $c = Q / (m \times \text{change in temperature})$.

Model answer: $c = 500 / (2 \times 5) = 50 \text{ J/(kg} \cdot \text{degree C)}$.

Write a well-organized, extended response of about 120-200 words.

91. Describe, step by step, how you would connect a battery, a switch, a bulb, and connecting wires to light the bulb in a simple series circuit. Then explain what would happen to the bulb's brightness if a second, identical bulb were added (a) in series and (b) in parallel.

Model answer / marking notes: A good answer describes connecting the components in a single closed loop (battery to switch to bulb and back to battery) so current can flow. Adding a second bulb in series increases the total resistance of the circuit, reducing the current, so both bulbs glow more dimly than a single bulb alone. Adding a second bulb in parallel keeps the voltage across each bulb equal to the battery voltage, so each bulb glows at roughly its normal brightness, while the total current drawn from the battery increases.

92. Explain the difference between real images and virtual images formed by lenses and mirrors, and give one real-world device that relies on each type of image.

Model answer / marking notes: A strong answer explains that a real image is formed where light rays actually converge and can be projected onto a screen (e.g., in a camera or a projector), while a virtual image is formed where light rays only appear to diverge from, and cannot be projected onto a screen (e.g., the image seen in a flat bathroom mirror or through a magnifying glass held close to an object).

93. Explain how a simple generator produces an electric current, referring to electromagnetic induction, and describe one everyday application of generators.

Model answer / marking notes: A strong answer explains that rotating a coil of wire within a magnetic field (or moving a magnet relative to a coil) continuously changes the magnetic flux passing through the coil, which by the principle of electromagnetic induction induces an electromotive force (voltage) that drives a current through the circuit. Everyday applications include bicycle dynamos, car alternators, and large-scale generators at hydroelectric, wind, or thermal power stations.

94. State the law of conservation of energy, and use it to explain what happens to the kinetic energy of a ball as it rises into the air, momentarily stops at its highest point, and then falls back down.

Model answer / marking notes: A strong answer states that energy cannot be created or destroyed, only transformed from one form to another (or transferred between objects). As the ball rises, its kinetic energy is progressively converted into gravitational potential energy; at the highest point its kinetic energy is momentarily zero while its potential energy is at a maximum; as it falls, potential energy converts back into kinetic energy, so the total mechanical energy stays constant throughout (ignoring small losses to air resistance).

Biology

Answer all questions in the spaces provided.

Choose the letter of the correct answer for each question.

1. Which organelle is known as the "powerhouse of the cell"?

- (A) Nucleus
- (B) Mitochondria
- (C) Ribosome
- (D) Golgi apparatus

Correct answer: (B) Mitochondria

Explanation: Mitochondria produce most of the cell's ATP through cellular respiration.

2. The genetic material of a eukaryotic cell is mainly found in the:

- (A) Cytoplasm
- (B) Nucleus
- (C) Cell membrane
- (D) Vacuole

Correct answer: (B) Nucleus

Explanation: DNA is housed within the nucleus of a eukaryotic cell.

3. In Mendelian genetics, an organism with two identical alleles for a trait is described as:

- (A) Heterozygous
- (B) Homozygous
- (C) Hybrid
- (D) Mutant

Correct answer: (B) Homozygous

Explanation: Homozygous means both alleles for that gene are the same (e.g., AA or aa).

4. Which chamber of the heart pumps oxygenated blood out to the rest of the body?

- (A) Right atrium
- (B) Right ventricle
- (C) Left atrium
- (D) Left ventricle

Correct answer: (D) Left ventricle

Explanation: The left ventricle has the thickest muscular wall and pumps oxygenated blood into the aorta and around the body.

5. Gas exchange in the lungs takes place in tiny air sacs called:

- (A) Bronchi
- (B) Trachea
- (C) Alveoli
- (D) Bronchioles

Correct answer: (C) Alveoli

Explanation: Alveoli are thin-walled, richly supplied with capillaries, giving a large surface area for gas exchange.

6. A group of different species living and interacting in the same area is called a:

- (A) Population
- (B) Community
- (C) Ecosystem
- (D) Biome

Correct answer: (B) Community

Explanation: A community consists of all the different populations (species) living and interacting in one area.

7. The process by which green plants make their own food using light energy is called:

- (A) Respiration
- (B) Photosynthesis
- (C) Transpiration
- (D) Fermentation

Correct answer: (B) Photosynthesis

Explanation: Photosynthesis converts light energy, water, and carbon dioxide into glucose and oxygen.

8. If both parents are heterozygous carriers (Aa) for a recessive trait, what fraction of their offspring is expected to show the recessive phenotype?

- (A) 1/4
- (B) 1/2
- (C) 3/4
- (D) 0

Correct answer: (A) 1/4

Explanation: Aa x Aa gives genotype ratio 1 AA : 2 Aa : 1 aa, so 1/4 of offspring are aa and show the recessive trait.

9. The double-helix structure of DNA is famously credited to:

- (A) Gregor Mendel
- (B) Charles Darwin
- (C) Watson and Crick
- (D) Louis Pasteur

Correct answer: (C) Watson and Crick

Explanation: James Watson and Francis Crick proposed the double-helix model of DNA structure in 1953.

10. Which nitrogenous base is found only in DNA, not in RNA?

- (A) Adenine
- (B) Guanine
- (C) Cytosine
- (D) Thymine

Correct answer: (D) Thymine

Explanation: DNA contains thymine, while RNA contains uracil in its place.

11. The process by which organisms with favorable inherited traits are more likely to survive and reproduce is called:

- (A) Mutation
- (B) Natural selection
- (C) Genetic drift
- (D) Cloning

Correct answer: (B) Natural selection

Explanation: This is the core mechanism of Darwin's theory of evolution by natural selection.

12. Which blood vessel carries deoxygenated blood away from the heart?

- (A) Aorta
- (B) Pulmonary vein
- (C) Pulmonary artery
- (D) Vena cava

Correct answer: (C) Pulmonary artery

Explanation: The pulmonary artery carries deoxygenated blood from the right ventricle to the lungs.

13. The type of cell division that produces gametes (sex cells) with half the normal number of chromosomes is called:

- (A) Mitosis
- (B) Meiosis
- (C) Binary fission
- (D) Budding

Correct answer: (B) Meiosis

Explanation: Meiosis reduces the chromosome number by half, producing genetically varied gametes.

14. In a food chain, organisms that make their own food using light energy are called:

- (A) Consumers
- (B) Decomposers
- (C) Producers
- (D) Predators

Correct answer: (C) Producers

Explanation: Producers (such as green plants and algae) make their own food through photosynthesis.

15. Which of the following is an example of a decomposer?

- (A) Lion
- (B) Oak tree
- (C) Mushroom (fungus)
- (D) Eagle

Correct answer: (C) Mushroom (fungus)

Explanation: Fungi such as mushrooms break down dead organic matter, making them decomposers.

16. A person with blood type O is generally considered a universal:

- (A) Donor
- (B) Recipient
- (C) Carrier
- (D) Mutant

Correct answer: (A) Donor

Explanation: Type O blood lacks A and B antigens, so it can generally be donated to people of any ABO blood type.

17. Compared with the air breathed in, the air breathed out by humans contains more:

- (A) Oxygen
- (B) Nitrogen
- (C) Carbon dioxide
- (D) Hydrogen

Correct answer: (C) Carbon dioxide

Explanation: Cellular respiration produces carbon dioxide, so exhaled air contains more CO₂ than inhaled air.

18. Which structure in a plant cell is mainly responsible for photosynthesis?

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Vacuole

Correct answer: (B) Chloroplast

Explanation: Chloroplasts contain chlorophyll and carry out photosynthesis.

19. The movement of water across a semi-permeable membrane from high to low water concentration is called:

- (A) Diffusion
- (B) Osmosis
- (C) Active transport
- (D) Transpiration

Correct answer: (B) Osmosis

Explanation: Osmosis specifically refers to the diffusion of water across a semi-permeable membrane.

20. A characteristic that is controlled by genes and passed from parent to offspring is called a(n):

- (A) Adaptation
- (B) Inherited trait
- (C) Acquired trait
- (D) Mutation event

Correct answer: (B) Inherited trait

Explanation: Inherited traits are passed genetically from parents to offspring, unlike acquired traits gained during an organism's lifetime.

21. Which organ is mainly responsible for filtering waste products out of the blood?

- (A) Liver
- (B) Kidney
- (C) Heart
- (D) Lung

Correct answer: (B) Kidney

Explanation: The kidneys filter blood and remove waste products, producing urine.

22. The scientific (binomial) name of an organism is typically written using:

- (A) One word
- (B) Two words (genus and species)
- (C) Three words
- (D) A numeric code

Correct answer: (B) Two words (genus and species)

Explanation: Binomial nomenclature gives each organism a two-part Latin name: genus followed by species.

23. The process by which plants lose water vapor through small pores called stomata is called:

- (A) photosynthesis
- (B) respiration
- (C) transpiration
- (D) germination

Correct answer: (C) transpiration

Explanation: Transpiration is the loss of water vapor from leaves via stomata.

24. Which blood cell type is primarily responsible for fighting infection?

- (A) Red blood cells
- (B) White blood cells
- (C) Platelets
- (D) Plasma

Correct answer: (B) White blood cells

Explanation: White blood cells are part of the immune system and fight pathogens.

25. Which is the most specific (narrowest) level of biological classification?

- (A) Kingdom
- (B) Genus
- (C) Species
- (D) Family

Correct answer: (C) Species

Explanation: Species is the most specific classification rank, below kingdom, family, and genus.

26. Which of these is NOT a function of the skeletal system?

- (A) Support
- (B) Protection of organs
- (C) Production of digestive enzymes
- (D) Movement (with muscles)

Correct answer: (C) Production of digestive enzymes

Explanation: Digestive enzymes are produced by digestive organs/glands, not the skeletal system.

Circle T if the statement is true, or F if it is false.

27. **T** Ribosomes are the site of protein synthesis in a cell.

28. **F** Mature human red blood cells contain a nucleus.

29. **F** Dominant alleles are always more common in a population than recessive alleles.

30. **T** Blood is carried away from the heart through arteries.

31. **F** Photosynthesis occurs in the mitochondria of plant cells.

32. **T** An ecosystem includes both the living organisms and the non-living environment in an area.

33. **T** Oxygen diffuses from the alveoli into the blood because its concentration is higher in the alveoli than in the blood.

34. **F** All mutations are harmful to the organism that carries them.

35. **T** DNA is composed of a double helix made of two strands of nucleotides.

36. **F** Meiosis produces four genetically identical daughter cells.

37. **T** Natural selection acts on genetic variation that already exists within a population.

38. **T** Decomposers break down dead organic matter and recycle nutrients back into an ecosystem.

39. **F** All bacteria are harmful to humans.

40. **T** Charles Darwin proposed the theory of evolution by natural selection.

41. **F** RNA contains the base thymine instead of uracil.

42. **T** Mitosis results in two daughter cells that are genetically identical to the parent cell.

43. **T** Osmosis is the movement of water across a semi-permeable membrane from high to low water concentration.

44. **F** Chloroplasts are found in animal cells as well as in plant cells.

45. **F** Acquired traits, such as a scar or a learned skill, can be passed on genetically to offspring.

46. **T** The kidneys play a major role in filtering waste products from the blood and producing urine.

47. **T** Binomial nomenclature names an organism using its genus and species.

48. **T** All living things are made up of at least one cell.

49. **T** Transpiration is the loss of water vapor from a plant's leaves through the stomata.
50. **T** White blood cells are primarily responsible for helping the body fight infection.
51. **F** Species is a broader classification category than kingdom.
52. **F** The skeletal system's main functions include producing digestive enzymes.

Fill in each blank with the correct word, phrase, or value.

53. The basic unit of heredity, passed from parent to offspring, is called a(n) **gene**.
54. The cell structure that controls what substances enter and leave the cell is the **cell membrane**.
55. An organism's observable physical characteristics are called its **phenotype**; **genotype**, while its genetic makeup is called its **phenotype**; **genotype**.
56. The muscular organ that pumps blood around the body is the **heart**.
57. Gas and nutrient exchange between blood and body tissues occurs across thin-walled vessels called **capillaries**.
58. A stable community of organisms together with their physical environment is called a(n) **ecosystem**.
59. Plants absorb the gas **carbon dioxide (CO₂)** from the air during photosynthesis.
60. For any gene, an offspring receives one allele from **each** parent.
61. The molecule that carries genetic instructions in living organisms is called **DNA**.
62. The type of cell division that produces two genetically identical daughter cells for growth and repair is called **mitosis**.
63. Organisms that break down dead organic material and recycle nutrients are called **decomposers**.
64. Darwin's theory explaining how species change over time through survival of the fittest is called **natural selection (evolution)**.
65. A change in an organism's DNA sequence is called a **mutation**.
66. The four nitrogenous bases found in DNA are adenine, thymine, cytosine, and **guanine**.
67. The natural environment in which an organism lives is called its **habitat**.
68. The scientific practice of grouping living things based on shared characteristics is called **classification (taxonomy)**.
69. The movement of water across a semi-permeable membrane from high to low water concentration is called **osmosis**.
70. The organelle mainly responsible for photosynthesis in plant cells is the **chloroplast**.
71. The organs that filter waste products from the blood and produce urine are the **kidneys**.
72. A trait passed from parent to offspring through genes, rather than gained during life, is called a(n) **inherited** trait.
73. The naming system that gives every organism a two-part Latin name is called **binomial** nomenclature.
74. The basic structural and functional unit of all living organisms is the **cell**.
75. The loss of water vapor from a plant's leaves through pores called stomata is called **transpiration**.
76. The type of blood cell mainly responsible for fighting infection is the **white** blood cell.
77. The most specific (narrowest) level of biological classification is the **species**.

78. Bones provide support and protection, and enable **movement** when combined with muscles.

79. Table 1 - Punnett square for a cross between two heterozygous parents (Aa x Aa)

	A	a
A	AA	Aa
a	Aa	aa

80. Table 2 - Organs and their main functions

Organ	Main function	Body system
Heart	Pumps blood around the body	Circulatory
Lungs	Gas exchange (breathing)	Respiratory
Kidney	Filters blood and removes waste	Excretory

81. Table 3 - Cell transport processes

Process	Requires energy (ATP)?	Direction of movement
Diffusion	No	High to low concentration
Osmosis (water)	No	High to low water concentration
Active transport	Yes	Low to high concentration

82. For each structure below, put a tick in the column that best describes where it is found.

Item	Plant cells only	Animal cells only	Both plant and animal cells
Cell wall	X		
Mitochondria			X
Chloroplast	X		
Nucleus			X
Large central vacuole	X		
Ribosomes			X
Cell membrane			X
Centrioles		X	

Answer in one to three complete sentences.

83. State two differences between arteries and veins.

Model answer: Any two of: arteries carry blood away from the heart, veins carry blood toward the heart; artery walls are thicker and more muscular than vein walls; arteries usually carry oxygenated blood (except the pulmonary artery) while veins usually carry deoxygenated blood (except the pulmonary vein); veins contain valves to prevent backflow of blood, arteries generally do not.

84. In pea plants, purple flower color (P) is dominant over white (p). A heterozygous plant (Pp) is crossed with a white-flowered plant (pp). Show the cross and state what fraction of the offspring is expected to have purple flowers.

Model answer: Pp x pp gives offspring Pp, Pp, pp, pp - so 1/2 (50%) are expected to have purple flowers (Pp) and 1/2 white (pp).

85. Explain why alveoli are well adapted for efficient gas exchange (state and explain two features).

Model answer: Any two of: a large surface area (many alveoli) increases the area available for diffusion; very thin, one-cell-thick walls shorten the diffusion distance; a moist inner surface helps gases dissolve; a rich network of capillaries maintains a steep concentration gradient by constantly supplying fresh blood.

86. Define "population" and "community" in ecological terms, and explain the difference between them.

Model answer: A population is all the individuals of the same species living in a given area at a given time. A community is all the different populations (of different species) living and interacting together in the same area. The difference is that a population involves one species, while a community involves many interacting species.

87. Explain what is meant by the term "homozygous recessive" and give an example genotype.

Model answer: It means an organism has two identical recessive alleles for a particular gene (e.g., genotype "aa"), and as a result it displays the recessive phenotype.

88. State one similarity and one difference between mitosis and meiosis.

Model answer: Similarity: both are types of cell division that begin from a single parent cell. Difference: mitosis produces two genetically identical diploid daughter cells (for growth and repair), while meiosis produces four genetically varied haploid daughter cells (for reproduction).

89. Explain, in one or two sentences, how antibiotic resistance in bacteria can develop through natural selection.

Model answer: Random mutations may make a few bacteria resistant to an antibiotic; when the antibiotic is used, only the resistant bacteria survive and reproduce, passing the resistance gene on, so the resistant population grows larger over successive generations.

90. Name the four nitrogenous bases found in DNA, and state which two always pair together.

Model answer: Adenine, thymine, cytosine, and guanine. Adenine pairs with thymine (A-T), and cytosine pairs with guanine (C-G).

Write a well-organized, extended response of about 120-200 words.

91. Describe the pathway of blood through the human heart and lungs in one complete circuit, starting from the right atrium, and explain why this double circulation system helps deliver oxygen efficiently to body tissues.

Model answer / marking notes: A complete answer traces: right atrium -> right ventricle -> pulmonary artery -> lungs (gas exchange, blood picks up oxygen and releases carbon dioxide) -> pulmonary vein -> left atrium -> left ventricle -> aorta -> body tissues -> vena cava -> back to the right atrium. Double circulation (separate pulmonary and systemic loops) keeps oxygenated and deoxygenated blood from mixing and allows blood to be re-pressurized by the left ventricle after returning from the lower-pressure lungs, so oxygenated blood reaches body tissues at higher pressure and more efficiently.

92. Using a simple food web of your choice, explain how energy flows through an ecosystem from producers to consumers, and describe what would likely happen to the rest of the food web if the primary producers were removed.

Model answer / marking notes: A good answer describes producers (e.g., grass or algae) capturing light energy through photosynthesis and passing chemical energy to primary consumers (herbivores) when eaten, then to secondary and tertiary consumers (carnivores), with energy lost as heat at each transfer (only about 10% passed on). If producers were removed, primary consumers would lose their food source and decline or die out, which would in turn remove the food source for the consumers that depend on them, potentially causing the collapse of the entire food web.

93. Explain the theory of evolution by natural selection in your own words, using an example (such as antibiotic-resistant bacteria, peppered moths, or another example of your choice) to illustrate how a population can change over many generations.

Model answer / marking notes: A strong answer explains that individuals within a population show natural variation; some variations give an individual a survival or reproductive advantage in its environment; individuals with advantageous traits are more likely to survive and reproduce, passing those traits to their offspring; and over many generations, the advantageous trait becomes more common in the population. A valid, clearly explained example (such as peppered moths becoming darker during industrial pollution, or bacteria evolving antibiotic resistance) earns full credit.

94. Explain the process of osmosis, and describe what would happen to a plant cell's turgor pressure if it were placed in a solution with a much higher solute concentration than its cytoplasm (a hypertonic solution).

Model answer / marking notes: A strong answer defines osmosis as the movement of water across a semi-permeable membrane from an area of high water concentration to an area of low water concentration. In a hypertonic solution, water would move out of the plant cell by osmosis; the cytoplasm would shrink away from the cell wall (plasmolysis), and the cell would lose turgor pressure, becoming flaccid.

English

Answer all questions in the spaces provided.

Reading Passage

The Rise of Renewable Energy in Southeast Asia

Over the past decade, renewable energy has grown faster in Southeast Asia than any other power source. Solar power, in particular, has expanded rapidly across countries such as Thailand, Vietnam, and the Philippines. Two main factors have driven this growth. First, the cost of manufacturing solar panels has fallen sharply, making solar electricity cheaper than ever before. Second, many governments in the region have introduced incentives, such as tax breaks and subsidies, to encourage both households and businesses to install solar systems.

Wind power has also seen significant growth, especially in coastal areas where strong, steady winds make large turbines practical. Engineers have installed new wind farms along several coastlines, adding thousands of megawatts of clean electricity to national grids.

Despite this progress, fossil fuels such as coal and natural gas still supply the majority of the region's electricity. Experts argue that continued investment in renewable technology, along with supportive government policy, will be necessary if Southeast Asian countries are to meet their long-term climate targets.

Choose the letter of the correct answer for each question.

1. Choose the correct verb form: "By next year, she ____ her studies."

- (A) will have completed
- (B) completing
- (C) completes
- (D) has completed

Correct answer: (A) will have completed

Explanation: This describes an action that will be finished before a future point in time, which requires the future perfect tense: will have completed.

2. Which word is closest in meaning to "abundant"?

- (A) Scarce
- (B) Plentiful
- (C) Expensive
- (D) Fragile

Correct answer: (B) Plentiful

Explanation: "Abundant" means existing in large quantities, which is closest to "plentiful".

3. Which sentence is correctly punctuated?

- (A) Its a beautiful day, isnt it
- (B) It's a beautiful day, isn't it?
- (C) Its' a beautiful day, isnt' it?
- (D) It is a beautiful day isnt it

Correct answer: (B) It's a beautiful day, isn't it?

Explanation: "It's" (it is) and "isn't" both need apostrophes, and the sentence is a question, so it needs a question mark.

4. Which sentence uses the passive voice correctly?

- (A) The cake was eaten by the children.
- (B) The children eating the cake.
- (C) The cake eat the children.
- (D) Eaten the cake children.

Correct answer: (A) The cake was eaten by the children.

Explanation: Passive voice puts the object of the action first, followed by a form of "to be" and the past participle: "The cake was eaten by the children."

5. According to the passage, what has helped drive the growth of solar power in the region?

- (A) Rising panel costs and fewer regulations
- (B) Falling panel costs and government incentives
- (C) A shortage of fossil fuels
- (D) A decrease in electricity demand

Correct answer: (B) Falling panel costs and government incentives

Explanation: The passage states that falling manufacturing costs and government incentives such as tax breaks and subsidies drove the growth.

6. Which word is an antonym (opposite) of "increase"?

- (A) Rise
- (B) Expand
- (C) Decrease
- (D) Grow

Correct answer: (C) Decrease

Explanation: "Decrease" means to become smaller, which is the opposite of "increase".

7. Which of the following is the correctly spelled word?

- (A) Recieve
- (B) Receive
- (C) Receeve
- (D) Receve

Correct answer: (B) Receive

Explanation: The correct spelling follows "i before e except after c" only partially here - the correct spelling is simply "receive".

8. Choose the conjunction that best completes the sentence: "She studied hard, ____ she passed the exam."

- (A) but
- (B) so
- (C) or
- (D) yet

Correct answer: (B) so

Explanation: "So" correctly shows the result of studying hard (cause and effect).

9. Choose the correctly formed plural of "child":

- (A) childs
- (B) children
- (C) childes
- (D) childrens

Correct answer: (B) children

Explanation: "Children" is the irregular plural form of "child".

10. Which sentence is written in the correct verb tense?

- (A) Yesterday, I go to the market.
- (B) Yesterday, I went to the market.
- (C) Yesterday, I going to the market.
- (D) Yesterday, I goes to the market.

Correct answer: (B) Yesterday, I went to the market.

Explanation: "Yesterday" signals the simple past tense, so "went" is correct.

11. Which word is a synonym for "happy"?

- (A) miserable
- (B) joyful
- (C) angry
- (D) tired

Correct answer: (B) joyful

Explanation: "Joyful" shares a similar meaning to "happy".

12. Choose the correct preposition: "The book is ____ the table."

- (A) on
- (B) in
- (C) at
- (D) by

Correct answer: (A) on

Explanation: "On" correctly describes a book resting on the surface of a table.

13. Which sentence contains a simile?

- (A) The moon is a bright lamp in the sky.
- (B) Her smile was as bright as the sun.
- (C) The sun smiled down on us.
- (D) The wind whispered through the trees.

Correct answer: (B) Her smile was as bright as the sun.

Explanation: A simile compares two things using "like" or "as"; "as bright as the sun" is a simile.

14. Which sentence is capitalized correctly?

- (A) i live in bangkok, thailand.
- (B) I live in Bangkok, Thailand.
- (C) I live in bangkok, Thailand.
- (D) I Live In Bangkok, Thailand.

Correct answer: (B) I live in Bangkok, Thailand.

Explanation: Sentences begin with a capital letter, and proper nouns (place names) are always capitalized.

15. Which of these words is spelled correctly?

- (A) definately
- (B) definitely
- (C) definitly
- (D) deffinitely

Correct answer: (B) definitely

Explanation: The correct spelling is "definitely."

16. Which sentence is a correctly formed question?

- (A) Where you are going?
- (B) Where are you going?
- (C) Where you going are?
- (D) You are going where?

Correct answer: (B) Where are you going?

Explanation: Questions in English typically invert the subject and auxiliary verb: "Where are you going?"

17. Which sentence uses correct subject-verb agreement?

- (A) The list of items are on the desk.
- (B) The list of items is on the desk.
- (C) The list of items be on the desk.
- (D) The list of items being on the desk.

Correct answer: (B) The list of items is on the desk.

Explanation: The subject is the singular noun "list," so it takes the singular verb "is," regardless of the plural noun "items" that follows it.

18. Which sentence uses the correct homophone?

- (A) Their going to the park.
- (B) They're going to the park.
- (C) There going to the park.
- (D) Theyre going to the park.

Correct answer: (B) They're going to the park.

Explanation: "They're" is the contraction of "they are," which fits this sentence.

19. Which sentence is an example of alliteration?

- (A) The big brown bear bounced happily.
- (B) The dog ran across the yard.
- (C) It was a dark and stormy night.
- (D) The sun rose over the mountains.

Correct answer: (A) The big brown bear bounced happily.

Explanation: Alliteration repeats the same initial consonant sound; "big brown bear bounced" repeats the "b" sound.

20. Choose the correct verb form: "Neither of the answers ____ correct."

- (A) are
- (B) is
- (C) were
- (D) be

Correct answer: (B) is

Explanation: "Neither" is treated as a singular subject in formal grammar, so it takes the singular verb "is."

21. In the sentence "I wanted to go, but it was raining," which word is the conjunction?

- (A) wanted
- (B) but
- (C) raining
- (D) go

Correct answer: (B) but

Explanation: "But" joins the two independent clauses in the sentence.

22. Which sentence contains an example of onomatopoeia?

- (A) The bees buzzed around the flowers.
- (B) The sky was blue.
- (C) She walked to school.
- (D) He read a book.

Correct answer: (A) The bees buzzed around the flowers.

Explanation: "Buzzed" imitates the sound the bees make, which is onomatopoeia.

23. Which sentence correctly uses an apostrophe to show singular possession?

- (A) The dogs bone.
- (B) The dog's bone.
- (C) The dogs' bone.
- (D) The doges bone.

Correct answer: (B) The dog's bone.

Explanation: An apostrophe followed by "s" shows singular possession: "the dog's bone."

24. Choose the correct comparative form: "This book is _____ than that one."

- (A) more interesting
- (B) interestinger
- (C) most interesting
- (D) interesting

Correct answer: (A) more interesting

Explanation: Longer adjectives form the comparative with "more," not an "-er" ending.

25. Which of these words is a collective noun?

- (A) team
- (B) player
- (C) game
- (D) score

Correct answer: (A) team

Explanation: "Team" refers to a group treated as a single unit, making it a collective noun.

26. Which sentence is written in the active voice?

- (A) The ball was thrown by Tom.
- (B) Tom threw the ball.
- (C) The ball is thrown.
- (D) Thrown was the ball.

Correct answer: (B) Tom threw the ball.

Explanation: In "Tom threw the ball," the subject (Tom) performs the action directly, which is active voice.

Circle T if the statement is true, or F if it is false.

27. **F** According to the passage, the cost of solar energy has risen sharply over the past decade.

28. **F** A noun is a word that describes an action.

29. **T** According to the passage, government incentives have supported the growth of renewable energy in the region.
30. **T** An adverb can modify a verb, an adjective, or another adverb.
31. **F** According to the passage, fossil fuels are described as the fastest-growing energy source in Southeast Asia.
32. **T** The sentence "Running quickly, the bus was caught by him" contains a dangling modifier.
33. **F** According to the passage, wind power has seen no growth in the region.
34. **T** A comma splice occurs when two independent clauses are joined only by a comma.
35. **T** "Children" is the correct plural form of "child."
36. **T** A simile makes a comparison using the words "like" or "as."
37. **T** A metaphor makes a direct comparison without using "like" or "as."
38. **F** The sentence "Me and him went to the store" is grammatically correct standard English.
39. **T** An adjective describes or modifies a noun.
40. **F** The past tense of "go" is "goed."
41. **T** Personification gives human qualities to non-human things or objects.
42. **F** A synonym is a word that means the opposite of another word.
43. **F** "They're," "their," and "there" are homophones that are spelled and used identically.
44. **T** Alliteration is the repetition of the same initial consonant sound in nearby words.
45. **T** The word "neither" is treated as a singular subject in formal English grammar.
46. **T** Onomatopoeia refers to words that imitate the sound they describe, such as "buzz" or "splash."
47. **F** A run-on sentence correctly joins two independent clauses without proper punctuation or conjunctions.
48. **F** The subject of an English sentence must always come before the verb.
49. **T** An apostrophe followed by "s" is used to show singular possession, as in "the dog's bone."
50. **F** The comparative form of "interesting" is "interestinger."
51. **T** A collective noun refers to a group treated as a single unit, such as "team" or "flock."
52. **F** The sentence "Tom threw the ball" is written in the passive voice.

Fill in each blank with the correct word, phrase, or value.

53. She **had finished** (finish) her homework before dinner yesterday.
54. Neither the teacher nor the students **were** (be) ready for the fire drill.
55. This is the book **that / which** I told you about.
56. If I **were** (be) you, I would apologize.
57. The opposite of "ancient" is **modern**.
58. A word that joins two clauses or sentences together is called a(n) **conjunction**.
59. The comparative form of the adjective "good" is **better**.

60. He is one of the students who **have** (have) finished the project.
61. The plural of "mouse" (the animal) is **mice**.
62. A word that describes a noun is called a(n) **adjective**.
63. A comparison using the words "like" or "as" is called a **simile**.
64. The past tense of the verb "write" is **wrote**.
65. A word that means the same, or nearly the same, as another word is called a **synonym**.
66. In "He quickly ran to the store," the word "quickly" is a(n) **adverb**.
67. The subject of the sentence "The dog barked loudly" is **the dog**.
68. A group of words that expresses a complete thought is called a **sentence**.
69. The repetition of the same initial consonant sound in a phrase, such as "big brown bear," is called **alliteration**.
70. A word that imitates the sound it describes, such as "buzz" or "splash," is called a(n) **onomatopoeia**.
71. Two independent clauses joined without proper punctuation or a conjunction form a(n) **run-on** sentence.
72. The words "their," "there," and "they're" are examples of **homophones** - words that sound alike but differ in meaning and spelling.
73. In formal grammar, the pronoun "neither" takes a **singular** verb.
74. A sentence fragment is a group of words that does not express a **complete** thought.
75. An apostrophe followed by "s," as in "the dog's bone," shows singular **possession**.
76. The comparative form of "interesting" is "**more** interesting."
77. A noun referring to a group treated as a single unit, such as "team" or "flock," is called a **collective** noun.
78. In "Tom threw the ball," the sentence is written in the **active** voice.

79. Table 1 - Identify the part of speech of the bracketed word in each sentence

Sentence	Word	Part of speech
She sings (beautifully).	beautifully	adverb
The (quick) fox jumped.	quick	adjective
He (runs) every morning.	runs	verb
(Happiness) is contagious.	Happiness	noun

80. Table 2 - Verb tense forms

Base form	Past simple	Past participle
go	went	gone
eat	ate	eaten
begin	began	begun

write	wrote	written
-------	-------	---------

81. Table 3 - Choose the correct homophone to complete each sentence

Sentence	Word choices	Correct word
I can't find ____ keys.	their / there / they're	their
____ going to be late.	their / there / they're	they're
Put the book over ____.	their / there / they're	there

82. For each word below, put a tick in the column showing its part of speech.

Item	Noun	Verb	Adjective
happiness	X		
run		X	
beautiful			X
honesty	X		
jump		X	
curious			X
kindness	X		
swim		X	
difficult			X

Answer in one to three complete sentences.

83. According to the passage, name two reasons for the growth of solar power mentioned in the text.

Model answer: The falling cost of manufacturing solar panels, and government incentives such as tax breaks and subsidies.

84. Rewrite the following sentence in the passive voice: "The engineers installed the new solar panels."

Model answer: "The new solar panels were installed by the engineers."

85. Combine the following two sentences using an appropriate conjunction: "It was raining. We went for a walk."

Model answer: Example: "Although it was raining, we went for a walk." (or "It was raining, but we went for a walk.")

86. Identify and correct the grammatical error in this sentence: "Each of the students have submitted their assignment."

Model answer: The error is subject-verb agreement: "each" is singular, so the verb should be "has", not "have". Corrected: "Each of the students has submitted their assignment."

87. In your own words, summarize the main idea of the passage in one or two sentences.

Model answer: Open-ended; a good summary notes that renewable energy, especially solar and wind power, is growing rapidly in Southeast Asia due to falling costs and supportive government policy, although fossil fuels still supply most of the region's electricity.

88. Write one sentence that uses a simile to describe a thunderstorm.

Model answer: Open-ended; any valid simile using "like" or "as" is acceptable, e.g., "The thunder roared like a lion across the sky."

89. Correct the grammar error in this sentence: "Me and my friend is going to the cinema."

Model answer: "My friend and I are going to the cinema."

90. Identify the adjectives in this sentence and the noun each one describes: "The curious cat explored the dark attic."

Model answer: "Curious" describes "cat," and "dark" describes "attic."

Write a well-organized, extended response of about 120-200 words.

91. Write a short paragraph (about 120-150 words) describing the advantages and disadvantages of renewable energy, based on the passage and your own knowledge.

Model answer / marking notes: Graded on a rubric: relevant content (advantages such as lower emissions and falling costs; disadvantages such as reliability/intermittency or high upfront installation costs), logical organization, and correct grammar and vocabulary.

92. Write a short persuasive letter (150-200 words) to your school principal proposing one environmental change your school should make, including at least two supporting reasons.

Model answer / marking notes: Graded on a rubric: appropriate letter format (greeting, body, closing), a clear proposal, at least two convincing supporting reasons, persuasive tone, and correct grammar and spelling.

93. Write a short descriptive paragraph (about 100-150 words) about a place that is special to you, using at least one simile or metaphor.

Model answer / marking notes: Graded on a rubric: vivid, specific descriptive detail; correct use of at least one simile or metaphor; logical organization; and correct grammar and spelling.

94. Write a short narrative paragraph (about 100-150 words) that includes at least one example of alliteration and one example of onomatopoeia. Underline or clearly mark both examples.

Model answer / marking notes: Graded on a rubric: coherent, engaging short narrative; a clearly identifiable, correctly used example of alliteration; a clearly identifiable, correctly used example of onomatopoeia; and correct grammar and spelling.